

$$L(\alpha, \beta) = \prod_{i=1}^N \frac{y_i^{\alpha-1} (1 - y_i)^{\beta-1}}{B(\alpha, \beta)}$$

$$\begin{aligned} \log L(\alpha, \beta) &= \sum_{i=1}^N \log \frac{y_i^{\alpha-1} (1 - y_i)^{\beta-1}}{B(\alpha, \beta)} \\ &= \sum_{i=1}^N [\log y_i^{\alpha-1} + \log(1 - y_i)^{\beta-1} - \log B(\alpha, \beta)] \\ &= \sum_{i=1}^N [(\alpha - 1) \log y_i + (\beta - 1) \log(1 - y_i) - \log B(\alpha, \beta)] \\ &= \sum_{i=1}^N [(\alpha - 1) \log y_i + (\beta - 1) \log(1 - y_i)] - N \log B(\alpha, \beta) \\ \log L(\alpha, \beta) &= (\alpha - 1) \sum_{i=1}^N \log y_i + (\beta - 1) \sum_{i=1}^N \log(1 - y_i) - N \log B(\alpha, \beta) \end{aligned}$$